

이 준 호 Alex Junho Lee

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Dept. of Mechanical Systems Engineering
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Education **Korea Advanced Institute of Science and Technology (KAIST)**

Ph.D., Civil and Environmental Engineering, 2023
- Robotics Program, Smart City Program
Bachelor of Engineering, Mechanical Engineering, Feb 2017
- Double Major in Business and Technology Management (BTM)

Field of Interests Visual Localization, Multimodal Sensor Fusion, SLAM, Place Recognition, Spatial AI

Publications Alex Junho Lee, Ayoung Kim, "Event-based Real-time Optical Flow Estimation". In Proceedings of the *IEEE International Conference on Control, Automation and Systems (ICCAS)*, 2017.

Alex Junho Lee, Younggun Cho, Sungho Yoon, Joowan Kim, Ayoung Kim, "ViViD: Vision for Visibility Dataset". In Proceedings of the *IEEE International Conference on Robotics and Automation (ICRA) Workshop: Dataset Generation and Benchmarking of SLAM Algorithms for Robotics and VR/AR, Best Paper*, 2019.

Alex Junho Lee, Ayoung Kim, "EventVLAD: Visual Place Recognition with Reconstructed Edges from Event Cameras". In Proceedings of the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.

Alex Junho Lee, Hyun Myung, "Natural Language Representation as Features for Place Recognition". In Proceedings of the *IEEE International Conference on Ubiquitous Robots (UR)*, 2022.

Alex Junho Lee, Younggun Cho, Young-sik Shin, Ayoung Kim, Hyun Myung, "ViViD++: Vision For Visibility Dataset". *IEEE Robotics and Automation Letters (RA-L)*, 7(3):6282–6289, 2022.

Alex Junho Lee, Younggun Cho, Hyun Myung, "Low-cost Thermal Mapping for Concrete Heat Monitoring". In Proceedings of the *IEEE International Conference on Robotics and Automation (ICRA) Workshop: Future of Construction: Build Faster, Better, Safer - Together with Robots*, 2022.

Alex Junho Lee, Hyungtae Lim, Minho Oh, Wonho Song, Hyun Myung, "Volumetric Vegetation Monitoring from LiDAR Scans with Ground Estimation". In Proceedings of the *IEEE International Conference on Control, Automation and Systems (ICCAS)*, pp. 1378–1379, 2022.

Sumin Hu, Yeeun Kim, Hyungtae Lim, Alex Junho Lee, Hyun Myung, "eCDT: Event Clustering for Simultaneous Feature Detection and Tracking". In Proceedings of the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 3808–3815, 2022.

Seungwon Song, Hyungtae Lim, Alex Junho Lee, Hyun Myung, "DynaVINS: A Visual-Inertial SLAM for Dynamic Environments". *IEEE Robotics and Automation Letters (RA-L)*, 7(4):11523–11530, 2022.

Alex Junho Lee, Seungwon Song, Hyungtae Lim, Woojoo Lee, Hyun Myung, "(LC)²: LiDAR-Camera Loop Constraints for Cross-Modal Place Recognition". *IEEE Robotics and Automation Letters (RA-L)*, 8(6):3589–3596, 2023.

Alex Junho Lee, Wonho Song, Byeongho Yu, Duckyu Choi, Christian Tirtawardhana, Hyun Myung, "Survey of Robotics Technologies for Civil Infrastructure Inspection". *Jour-*

nal of Infrastructure Intelligence and Resilience, 2(1):100018, 2023.

Ung Hyun Ko, Jinseung Choung, Alex Junho Lee, Suk-Hee Park, Jennifer H. Shin, “Surface Tension-Induced Biomimetic Assembly of Cell-Laden Fibrous Bundle Construct for Muscle Tissue Engineering”. *Biomedical Materials*, 18(5):055031, 2023.

Seungwon Song, Hyungtae Lim, Alex Junho Lee, Hyun Myung, “DynaVINS++: Robust Visual-Inertial State Estimator in Dynamic Environments by Adaptive Truncated Least Squares and Stable State Recovery”. *IEEE Robotics and Automation Letters (RA-L)*, 9(10):9127–9134, 2024.

Seoyeon Jang, Alex Junho Lee, I Made Aswin Nahrendra, Hyun Myung, “Chamelion: Reliable Change Detection for Long-Term LiDAR Mapping in Transient Environments”. *IEEE Robotics and Automation Letters (RA-L)*, 11(4):4361–4368, 2026.

Alex Junho Lee, Dongjin Hyun, “언어 기반 네비게이션에서의 불확실성 재고”. *Journal of Korea Robotics Society (JKRS)*, 21(2):158–163, 2026.

Chaehyun Yoon, Alex Junho Lee, “Cross-Modal Optimization on the Special Linear Group for Camera-Based Localization and Mapping in 2D Occupancy Grids”. *International Journal of Control, Automation, and Systems (IJCAS)*, 2026.

Patents

Heeju Cho, Hyeseung Ahn, Soyeon Jeong, Jinhwa Hwang, Saeyeon Park, Areum Lee, Alex Junho Lee, “Method and Apparatus for Providing a Door Control Algorithm to Synchronously Open and Align a Pair of Doors Using a Linkage Mechanism in an Emergency,” Korean Patent Application No. 10-2026-0060451, 2026.

Heeju Cho, Hyeseung Ahn, Soyeon Jeong, Jinhwa Hwang, Saeyeon Park, Areum Lee, Alex Junho Lee, “Variable Revolving Door Apparatus Convertible from a Rotating Structure to a Linear Open-Passage Structure in an Emergency and Control Method Thereof,” Korean Patent Application No. 10-2026-0060452, 2026.

Ayoung Kim, Alex Junho Lee, Younggun Cho, Giseop Kim, Hyunchul Roh, “Encoder Frame Device and Vehicle Odometry Measurement System Using the Same,” Korean Patent Application No. 10-2017-0099614, Korean Patent Registration No. 10-1994339-0000, 2019.

Alex Junho Lee, “Device and Method for Generating Descriptors for Structural Landmarks in Geometrically Degenerate Environments,” Korean Patent Application No. 10-2026-0185198, 2026.

Articles

Modern SLAM and Spatial AI Technologies, *Robot and Human*, 23(1), Summer 2026.

Trends in Safety Robot Technologies for Building Inspection, *ICROS Magazine*, September 2022.

Technology Innovation Committee Report on Intelligent Robotics, 2021.

Fourth Industrial Revolution and Autonomous Driving Report, 2017.

Talks & Seminars

“Research Trends in Robust Perception Methods for Mobile Robots in Real-World Environments”
Korea Institute for Robot Industry Advancement (KIRIA), Aug. 19, 2026.

“Spatio-Temporal State Estimation and Semantic Memory in Unstructured Environments”
Korea Robotics Society Annual Conference (KRoC), Feb. 6, 2026.

“Spatial AI Tutorial - AI Fundamentals for Monocular VSLAM”
Korea Robotics Society Annual Conference (KRoC), Feb. 6, 2026.

“Map-Independent Spatial Perception in Unstructured Extreme Environments”
Korea Institute of Machinery & Materials (KIMM), Nov. 10, 2025.

“Map Quality Evaluation for Robotics”
International Conference on Control, Automation, and Systems (ICCAS), Nov. 6, 2025.

“Surviving as a Mechanical Engineer in the AI Era”
Graduate Seminar, Sookmyung Women’s University, Oct. 13, 2025.

“Beyond SLAM: Distributed Sensors, Collective Data”
10th Anniversary Symposium, College of Engineering, Sookmyung Women’s University, Sep. 6, 2025.

“Conducting Robot Perception Research as an Early-Career Researcher”
Institute of Advanced Materials and Systems, Sookmyung Women’s University, Aug. 29, 2025.

“What Is Spatial Intelligence for Mobile Robots?”
AI Center, Sookmyung Women’s University, May 23, 2025.

Invited Talk
Pusan National University, Nov. 25, 2024.

“Multimodal Spatial Intelligence for Robots”
Department of Electrical Engineering, Inha University, Nov. 21, 2024.

“Service Robot Operation in Real-World Environments and Multimodal Spatial Intelligence”
Young Researchers Session, Institute of Control, Robotics and Systems (ICROS), Jul. 3, 2024.

Invited Talk
Ajou University, Apr. 18, 2024.

Invited Talk
DGIST, Jan. 15, 2024.

Invited Talk
Pusan National University, Nov. 26, 2023.

“Foundation Models in Robotic Missions”
SLAM KR, Nov. 25, 2023.

“Robotic Foundation Models”
Department of Electrical Engineering, Inha University, Nov. 21, 2023.

“Robust sensor fusion-based localization”
Korea Electronics Technology Institute (KETI), Oct. 26, 2023.

Research Projects Google Cloud Education Grant (USD 5,000), Google, 2026.

Development of Exploration Methods for Precise 3D Reconstruction in Unknown Complex Environments, Sookmyung Women’s University, 2026.

Integration of Localization System and Software for Sewer Pipeline Mapping System Development, EPS E&E Co., Ltd., 2026.

Research on Localization Methods for Autonomous Robot Operation in Large-Scale Environments, Sookmyung Women's University, 2025.

Creative Asset Commercialization Support Program (BRIDGE 3.0), Industry-Academic Cooperation Foundation, Sookmyung Women's University, 2025.

Awards

Top Excellence Award (KRW 1,000,000), 2025 SMU EDU Expo
Capstone Design Competition — Supervisor, Team MechX, Dec. 2025.

Top Excellence Award (KRW 1,000,000), 2025 SMU EDU Expo
Capstone Design Competition — Supervisor, Team Combeeneration, Dec. 2025.

Outstanding Young Researcher,
International Conference on Control, Automation, and Systems (ICCAS), 2025.

Outstanding Early Career Researcher,
Institute of Control, Robotics and Systems (ICROS), 2024.

2nd Cash Award, *HILTI SLAM CHALLENGE*, 2022.

Best Paper, *IEEE Int. Conf. Robotics and Automation (ICRA) Workshop: Dataset Generation and Benchmarking of SLAM Algorithms for Robotics and VR/AR*, 2019.

Experiences

Robust visual place recognition for location authentication (Project, 2022)
- Led the project and developed a deep-learning-based visual place recognition system.

Last-mile delivery robot in crowded urban areas (Project, 2021–2022)
- Developed LiDAR-based SLAM for an unmanned ground vehicle as the project's SLAM engineer.

K-DARPA Challenge (Final Stage, 2021)
- Developed pose and object-location estimation for drone flight tests in battlefield environments.
- Ran stereo VIO and a tilted 2D LiDAR on an embedded device (Jetson TX2) to recognize objects of specified shapes and map their locations.

Future directions of robotics research (Project, 2021)
- Contributed to a study of future robotics research directions.

Outdoor SLAM in unstructured environments (Project, 2019–2021)
- Led the development of autonomous mapping for construction sites.
- Active SLAM, Long-term mapping, Sensor Integration

Indoor SLAM with dynamic obstacles (Project, 2019)
- Developed the SLAM component of a general-purpose indoor service robot.
- Implemented SLAM and obstacle avoidance for dynamic indoor environments.

Encoder frame device and vehicle odometry measurement system (Patent, 2019)
- Proposed and built a high-resolution encoder frame for vehicle odometry.

Indoor SLAM under compound disaster conditions (Project, 2018)
- Developed localization and mapping for a rescue drone in compound-disaster scenarios such as earthquakes and fires.

Fourth Industrial Revolution and autonomous driving (Project, 2017)
- Contributed to a study of policy requirements for autonomous driving.

Intern, Intelligent Robotics Research Division (Electronics and Telecommunications Research Institute, 2017)

Intern, Safety Device Design Team (Doosan Heavy Industries, 2016)

International student exchange program (National University of Singapore, 2016)

Educational volunteer (Daejeon Yuseong-gu, 2013–2015)

Teaching

Mechanism Design (Spring 2025; Spring 2026)

Engineering Mathematics (Spring 2026)

CAD & 3D Printing (Fall 2025)

Graduate Numerical Analysis (Fall 2025)

Service

Associate Editor: IEEE Robotics and Automation Letters (RA-L), 2026–Present; IEEE International Conference on Robotics and Automation (ICRA).

Editor: IEEE International Conference on Ubiquitous Robots (UR), 2025–2026.

Guest Editor: International Journal of Control, Automation, and Systems (IJCAS), 2025–2026.

Session Chair: Session Co-Chair, Localization II — IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Prague, 2021.

Reviewer: IEEE T-RO, T-IV, T-ASE, RA-L, RAS, Neurocomputing, ICRA, IROS, ICCAS, UR, IJCAS, JIST.

Instructor: KAIST Global Center for Gifted Children, Mar. 2015–Dec. 2018.

Language

Korean (Native), English (Fluent)

C++, Python